

Improving Evidence Retrieval for Scientific QA via Rich Contextualization and Domain Adaptation

NLP Project Final Report

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Abstract

Automated evidence retrieval in document-level scientific question answering (QA) is a challenging NLP problem due to long-context reasoning requirements and domain-specific language. We target three gaps in the PeerQA benchmark [1]: query expansion, domain-specific fine-tuning of dense retrievers, and richer structural contextualization. By addressing the lexical gap between reviewer language and paper terminology and the token-overwhelming problem that limits sentence-level retrieval, we evaluate three retrieval approaches on the full 383-question PeerQA evaluation set. Query2Doc augmentation of BM25 reaches MRR = 0.4432, surpassing the published PeerQA BM25 baseline of 0.4288 (+15.1% relative over our own BM25 baseline). Structural context enrichment achieves MRR = 0.4036 at paragraph level using the full_rich format at zero API cost, with abstract_only as the dominant single component. A supervised contrastive fine-tuning of MiniLM (3-fold CV, 1 epoch) achieves MRR = 0.3618, outperforming the published zero-shot Contriever baseline (0.3494). Downstream generation with the best retrieval system yields Rouge-L F1 = 0.1686, confirming that retrieval quality is the primary bottleneck for scientific QA.

CCS Concepts

• Information systems → Question answering.

Keywords

Scientific QA, RAG, PeerQA, Domain Adaptation, Evidence Retrieval, BM25, Dense Retrieval

1 Project Idea

The central problem is automated evidence retrieval in document-level scientific QA. Given a question from a peer review and a paper averaging over 11,700 tokens, a system should be able to identify the exact paragraph or sentence containing the answer. So this is what makes it not just a simple keyword search. This will involve NLP challenges like long-context reasoning. In a Retrieval-Augmented Generation (RAG) [10] pipeline, the generator’s quality is constrained by the retriever’s selection. PeerQA [1] shows a correlation between retrieval recall and generative correctness as high as $r = 0.42$. Improving retrieval is probably the highest-leverage intervention for scientific QA. We plan to target three gaps: query expansion, domain-specific fine-tuning of dense retrievers, and richer structural contextualization. This requires understanding information retrieval, embedding models, and the nature of scientific text.

2 Motivation

The number of scientific literature is increasing every year, so this is creating a burden for researchers. An automated QA can accelerate verification and synthesis. So PeerQA tested models strictly in zero-shot capacities. No model was fine-tuned for scientific text, and title prepending failed at the sentence level due to token overwhelming. Reviewers used informal shorthand (e.g., “this approach”), while papers use precise technical terms; so this lexical gap motivates query expansion and domain adaptation. Identifying specific sentences reduces hallucination risk. Sentence-level MRR (0.36) lags behind paragraph-level (0.48), creating some room for improvement. We hypothesize that richer context (section titles, paragraph position) creates semantic anchors that orients the retriever without dominating the sentence’s signal.

3 Dataset

We will use the PeerQA dataset [1], released in February 2025 by Baumgartner, Briscoe, and Gurevych from the UKP Lab at TU Darmstadt. PeerQA received the Outstanding Paper Award at NAACL 2025 and is publicly available at <https://github.com/UKPLab/peerqa> and on Hugging Face at UKPLab/PeerQA [5].

3.1 Dataset Construction

PeerQA was constructed through a rigorous multi-stage pipeline designed to produce realistic, expert-quality scientific QA data. First, 17,910 candidate questions were extracted from peer reviews of papers submitted to ML/NLP venues (ICLR 2022 and 2023, NeurIPS 2022, ARR 2022, ACL, COLING, CoNLL) and geoscience journals (Earth System Dynamics, Earth Surface Dynamics), plus F1000. Questions were decontextualized using InstructGPT to make them self-contained, decomposed if compound, and filtered manually to retain only information-seeking questions about paper content. The final set of processed questions was then sent to the original paper authors for annotation. Authors highlighted evidence sentences in the camera-ready PDF and provided free-form text answers. This expert-author annotation loop is the defining quality advantage of PeerQA over all prior scientific QA datasets.

3.2 Key Statistics

3.3 Why PeerQA is Appropriate

PeerQA [1] is the ideal dataset for this project for several reasons. First, its questions are natural: they were actually raised by expert reviewers during the peer review process, not artificially constructed by crowdworkers. This makes the dataset the most realistic

Table 1: PeerQA Dataset Statistics

Attribute	Value
Total annotated QA pairs	579
Academic papers covered	208
Avg. paper length	~11,700 (Llama-3 [4] tokens)
Avg. question length	20.2 tokens
Answerable questions	383
Unanswerable questions	112
Non-consecutive evidence	~30%
Unlabeled questions	11,967 from 2,623 papers
Primary domains	ML, NLP, Geoscience, Public Health

available benchmark for scientific QA. Second, its annotations are by the paper authors themselves, the highest possible domain authority, ensuring that the evidence labels are definitive rather than approximate. Third, the dataset is not on Kaggle and has no existing student-project solutions online, meaning our work will be genuinely original. Fourth, the presence of 11,967 unlabeled questions enables an unsupervised domain pre-training strategy that would not be possible with smaller datasets. Finally, the paper publishes exact baseline numbers for MRR and Recall@10 across multiple retriever architectures, giving us clear benchmarks against which to measure our improvements.

Table 2: Dataset Comparison

Dataset	Annotator	Question Source	Scope
BioASQ	Experts	Expert-generated	Abstract
QASPER [3]	Practitioners	Abstract readers	Full paper
QASA [9]	Practitioners	Full-paper readers	Full paper
PeerQA [1]	Authors	Real peer reviews	Full paper

4 Examples from the Dataset

The following is a few of the examples of the data we will be analyzing. They illustrate the structure and challenge level of PeerQA retrieval. Each example has a question, the full paper as the search corpus and one or more gold evidence paragraphs. The retrieval task is to rank the gold paragraph(s) as high as possible across all paragraphs in the paper.

Table 3: Example 1 (Methods Clarification)

Field	Content
Question	Can the authors provide an explanation for the significant performance difference between CGExpan and LM-base?
Gold Evidence	SOTA method CGExpan outperforms other methods on existing well-curated benchmarks which aligns [...] CGExpan has an assumption that each entity should [...]
Retrieval Challenge	The correct paragraph must be found among ~80+ candidate paragraphs spanning 12,000 tokens. The question uses informal reviewer phrasing; the evidence uses formal paper terminology.

Table 4: Example 2 (Analysis)

Field	Content
Question	What impact does the layer norm approximation have on the performance of the tasks discussed in the paper?
Gold Evidence	To prevent the overflow attention scores, we use the weight decay of Adam optimizer as regularization [...] before normalization, we also observe the attention scores are scarily sparse, with some extreme values reaching $1e4$ [...]
Retrieval Challenge	The answer requires domain knowledge to connect “layer norm approximation” to “attention overflow” and “downstream task performance” — a semantic gap that defeats simple keyword matching.

Table 5: Example 3 (Geoscience — Multi-Location Evidence)

Field	Content
Question	Does the orientation of the grid, with potentially more north-facing slopes in the Elwha area, affect the absorption of solar radiation, thereby influencing glacier distribution?
Gold Evidence	Two non-consecutive paragraphs — one discussing the flowline model’s assumptions, one discussing topographic shading as a mass balance factor.
Retrieval Challenge	Evidence is split across two non-consecutive locations in the paper (30% of PeerQA questions have this property), requiring the retriever to rank multiple passages highly.

5 NLP Approaches

5.1 Approach 1: LLM-Based Query Rewriting (Query2Doc)

Starting with the first approach we will be trying is addressing the lexical gap. The PeerQA authors tested BM25 [12] as a baseline but

applied no query changes or augmentation. They used Standard synonym expansion using WordNet or spaCy and we think this is insufficient for highly technical scientific vocabulary. For example, terms like “constituency parser” or “inverse cloze task” have no WordNet synonyms. Instead, in our approach we implement LLM-based query expansion via the Query2Doc method [13]: given the reviewer question, we prompt gpt-4o to generate a short passage from a scientific paper that would directly contain the answer. The augmented query is $(question \times 5) + pseudo_document$; the $5\times$ repetition prevents the longer pseudo-document from drowning out original query terms in BM25 scoring [13]. We compare three configurations: BM25 original, BM25 with Query2Doc, and Reciprocal Rank Fusion (RRF) of the two.

An initial attempt used simple paraphrasing (“rewrite in formal academic terminology”), which consistently underperformed the BM25 baseline. LLM paraphrasers substitute synonyms while dropping technical terms—the worst transformation for a lexical bag-of-words matcher. Query2Doc inverts this: instead of rewriting the query, it generates the answer. The pseudo-document inherits formal scientific vocabulary, providing the exact term overlap BM25 needs.

Implementation: We use bm25s (v0.3.8) with lowercase + Porter stemmer + English stopword tokenization, matching the Pyserini Lucene EnglishAnalyzer used in the published PeerQA baseline. GPT-4o is used with temperature 0.3 and max 300 output tokens. Total API cost for 383 queries was approximately \$1.10 USD, with all pseudo-documents cached to eliminate re-run costs.

5.2 Approach 2: Domain-Adapted Dense Retriever

For the second approach we address the zero-shot limitation of all dense retrievers tested in PeerQA. Models like Contriever [6] and MiniLM were trained on general web text and Wikipedia, without exposure to the question-evidence structure of scientific papers.

We adapt all-MiniLM-L6-v2 to the scientific domain using supervised fine-tuning with MultipleNegativesRankingLoss—a standard in-batch negatives contrastive setup [7] where each (question, gold paragraph) pair is a positive training example and other in-batch passages serve as implicit negatives. For each training question, every gold paragraph is paired with one randomly sampled paragraph from the same paper as a hard negative, which forces the model to learn fine-grained intra-document discrimination rather than coarse domain boundaries. We use 3-fold cross-validation (seed = 42) to produce reliable performance estimates given the small labeled set. Inverse Cloze Task (ICT) pre-training was skipped due to CPU constraints.

Retrieval: Both queries and candidates are embedded to unit-norm 384-d vectors and are ranked by dot product (equivalent to cosine similarity).

Hyperparameters: batch size 16, 1 epoch, 10% warmup steps, AdamW optimizer, seed 42. Each fold trains a fresh model from pretrained weights. Fold sizes: $127 + 127 + 129 = 383$ validation questions.

5.3 Approach 3: Richer Structural Contextualization

In our last approach we directly target the sentence-level retrieval failure identified in PeerQA [1]. The current best sentence-level MRR (0.36) lags behind paragraph-level (0.48) by 0.118 points. The PeerQA authors attribute this partly to token overwhelming; the paper title is too semantically dominant relative to the short sentence.

Our hypothesis is that replacing the single title with a richer multi-component string provides semantic anchors without overwhelming the evidence signal. We test the following seven indexing templates (query stays raw; only document strings change):

- plain: sentence/paragraph text only
- title_only: Title | text (replicates PeerQA published baseline [14])
- title+section: title | heading | text
- title+section+pos: adds Para p Sent s position token
- abstract_only: abstract_summary | text (first sentence of abstract)
- full_rich: abstract_summary | title | heading | Para p (Sent s) | text
- section_only: heading | text

A fresh BM25 index is built per question over that paper’s candidates.

6 Evaluation Metrics

6.1 Phase 1: Retrieval Evaluation

Mean Reciprocal Rank (MRR): MRR measures the rank of the first correctly retrieved evidence passage. If the correct passage is ranked 1st, then the MRR = 1.0; if it is 2nd, MRR = 0.5; and 3rd would be MRR = 0.33, etc. MRR is the primary metric for us because in the real-world, a user or generator will probably focus on the top-ranked result. In the PeerQA paper the baselines are paragraph+title best = 0.4845 (Dragon+ [11]) and sentence+title best = 0.3654 (MiniLM). We will compare to these baselines.

Recall@10: Recall@10 is a metric that measures whether the correct evidence passage appears anywhere in the top 10 results. This is important for the RAG pipeline since if the correct passage is not in the top 10, it will never reach the generator. Recall@10 is used with MRR because a system can have high MRR but also a low Recall@10.

Reporting: For reporting we report both metrics at the paragraph level and sentence level, with and without our contextualization strategies. This allows us to compare directly against Table 3 of the PeerQA paper.

6.2 Phase 2: Downstream Generation

Retrieval improvements are only good if they produce better answers so we take the top-10 passages retrieved by our best-performing approach (RAG-10) and feed them as context to gpt-4o-mini (used as a CPU-accessible proxy for Llama-3-8B-Instruct [4]) to generate free-form answers.

Rouge-L: This metric measures the longest common subsequence between the generated answer and the expert-written reference answer and it is normalized by the answer length. Rouge-L captures

lexical overlap but it can penalize correct answers that use different phrasing. So in this case we use it alongside other metrics as well. **AlignScore [15]:** This is a model-based factual consistency metric trained on text alignment data. AlignScore checks whether each sentence in the generated answer is supported by information in the retrieved evidence paragraphs. This is important for scientific QA where hallucination is the main cause of failure. This metric is different from Rouge-L since it rewards correct answers in its semantics.

Success Criterion: We define success as achieving a statistically significant improvement in paragraph-level MRR or Recall@10 over the baselines. The strongest PeerQA baseline (Dragon+ [11] at 0.4845 MRR), or a meaningful improvement in sentence-level MRR.

7 Experimental Setup

All three approaches are evaluated on the full standard PeerQA evaluation set of **383 answerable questions** with mapped evidence paragraphs, spanning 167 unique papers and 28,496 paragraph chunks (average 137 paragraphs per paper). This matches the evaluation protocol of the published PeerQA paper [1] and enables direct comparison against published baselines.

For all BM25 experiments we use bm25s (v0.3.8) with lowercase + Porter stemmer + English stopword tokenization, matching the Pyserini Lucene EnglishAnalyzer configuration used in the published PeerQA baseline. Retrieval is performed per-paper, where each question is ranked against only the paragraphs of its source paper. For the dense retrieval (Approach 2) we use sentence-transformers [11] as our base model. Approach 2 was run with `-skip-ict` due to CPU constraints, and training was capped at 1 epoch per fold. All experiments are run on CPU.

8 Results

8.1 Approach 1: Query2Doc-Augmented BM25

Our initial implementation used a paraphrase-based rewriting prompt—asking the LLM to “rephrase this question in formal academic terminology.” These rewrites consistently underperformed the original BM25 baseline upon inspection, revealing the LLM substituted synonyms while dropping technical terms. For BM25, removing exact-term overlap with paper vocabulary without compensating with stronger keywords is counter-productive.

We therefore switched to **Query2Doc [13]**: instead of rewriting the *query*, we prompt gpt-4o to generate a short hypothetical *answer passage* as if written by the paper’s authors. The augmented query is constructed as $(question \times 5) + pseudo-doc$, where the $5\times$ repetition prevents the longer pseudo-document from drowning out the original query terms in BM25 scoring. Total API cost for all 383 questions was approximately \$1.10 USD, with pseudo-documents cached to avoid re-generation.

Table 6 reports the full results.

Table 6: Approach 1 — BM25 with Query2Doc Augmentation (n=383)

Method	MRR	R@1	R@5	R@10	R@20
BM25 (original)	0.3852	0.1929	0.4785	0.6072	0.7467
BM25 (RRF fused)	0.4174	0.2278	0.4913	0.6323	0.7674
BM25 + Query2Doc	0.4432	0.2500	0.5086	0.6527	0.7808
<i>PeerQA BM25 (published)</i>	0.4288	—	—	0.6817	—

Query2Doc achieves MRR = 0.4432, surpassing both our own BM25 baseline (+15.1% relative) and the published PeerQA BM25 baseline (0.4288). The largest relative gain is at R@1 (+29.6%: 0.1929 $\rightarrow$ 0.2500), which is the metric most directly relevant to a downstream RAG pipeline that feeds only the top-ranked paragraph to the generator.

Counterintuitively, Reciprocal Rank Fusion (RRF) of the original and Query2Doc rankings *hurts* relative to Query2Doc alone (MRR 0.4174 vs. 0.4432). This is a well-known RRF failure mode [2]: when component rankers are imbalanced, fusion averages the strong ranker back toward the weaker one. With Query2Doc at MRR 0.4432 and original BM25 at 0.3852, the fusion pulls performance down by ≈ 0.026 .

8.2 Approach 2: Supervised Dense Retriever

We fine-tune all-MiniLM-L6-v2 on the 383 labeled PeerQA pairs using MultipleNegativesRankingLoss with same-paper hard negatives. For each training question, every gold paragraph is paired with one randomly sampled paragraph from the same paper as a hard negative. We use 3-fold cross-validation (seed = 42), training for 1 epoch per fold (batch size 16, AdamW, 10% warmup steps). Table 7 reports the results; Table 8 shows the per-fold breakdown.

Table 7: Approach 2 — Zero-Shot vs. Supervised MiniLM (n=383)

Method	MRR	R@1	R@5	R@10	R@20
Zero-shot MiniLM	0.3286	0.1570	0.3766	0.5360	0.6929
Supervised (3-fold avg)	0.3618	0.1722	0.4207	0.5754	0.7362
<i>Contriever zero-shot (published)</i>	0.3494	—	—	—	—
<i>Dragon+ (published)</i>	0.4845	—	—	0.6817	—

Table 8: Approach 2 — Per-Fold Breakdown

Fold	Train Pairs	Val Qs	MRR	R@1	R@5
1	442	127	0.3518	0.1606	0.4354
2	437	127	0.3893	0.2023	0.4099
3	423	129	0.3444	0.1535	0.4169
Avg	—	—	0.3618	0.1722	0.4207
Std	—	—	± 0.0196	—	—

The fine-tuned model (MRR 0.3618) outperforms the published zero-shot Contriever baseline (0.3494)—a substantially larger model—demonstrating that limited in-domain supervision on a lightweight

encoder can outperform a larger zero-shot retriever. The largest relative gain over zero-shot MiniLM is at R@5 (+11.7%), indicating the model systematically pushes relevant paragraphs into the top-5 range most useful for a RAG generator. Cross-fold variance (MRR std ± 0.0196) is moderate and expected given the small labeled set (~440 training pairs per fold); all three folds consistently exceed the zero-shot baseline, confirming a reliable positive signal from domain supervision. Fold 2 is notably stronger (0.3893), likely reflecting random variation in held-out question difficulty. The training loss (1.59–1.69) indicates the model has not fully converged, suggesting that additional epochs or GPU-scale training would yield further improvement.

8.3 Approach 3: Richer Structural Contextualization

Here, we test seven context-string templates at both paragraph and sentence level under the same BM25 setup as Approach 1. The `abstract_summary` field is the first sentence of the abstract, extracted deterministically via NLTK `sent_tokenize` applied to the concatenated abstract-section chunks. Sentence units are recovered by the same method applied to each paragraph chunk, yielding 76,308 sentence units across 208 papers. Sentence-level retrieval is evaluated against paragraph-membership gold labels (a retrieved sentence counts as correct if its parent paragraph is in `relevant_chunk_ids`), so sentence-level absolute numbers are not directly comparable to PeerQA’s published per-sentence MRR; cross-format deltas within our pipeline remain valid. This approach has **zero API cost** and runs end-to-end in approximately 50 seconds on CPU.

As a sanity check, `plain` in Approach 3 produces MRR = 0.3852, identical to the bare BM25 baseline in Approach 1, verifying the two pipelines agree when no context is added.

Tables 9 and 10 report all results.

Table 9: Approach 3 — Paragraph-Level Results (n=383)

Format	MRR	R@1	R@5	R@10	R@20
<code>plain</code>	0.3852	0.1929	0.4785	0.6072	0.7467
<code>title_only*</code>	0.3873	0.1981	0.4888	0.6016	0.7237
<code>title+section</code>	0.3895	0.2001	0.4870	0.6162	0.7333
<code>title+section+pos</code>	0.3919	0.2028	0.4931	0.6201	0.7379
<code>section_only</code>	0.3896	0.1990	0.4812	0.6148	0.7477
<code>abstract_only</code>	0.4027	0.2171	0.4780	0.6235	0.7530
<code>full_rich</code>	0.4036	0.2172	0.4916	0.6164	0.7373
<i>PeerQA BM25 (published)</i>	0.4288	—	—	0.6817	—

*Replicates the PeerQA published BM25 template.

Table 10: Approach 3 — Sentence-Level Results (n=383)[†]

Format	MRR	R@1	R@5	R@10	R@20
<code>plain</code>	0.3509	0.2115	0.5013	0.6762	0.7937
<code>title_only*</code>	0.3536	0.2245	0.4856	0.6475	0.7885
<code>title+section</code>	0.3535	0.2193	0.5013	0.6554	0.7650
<code>title+section+pos</code>	0.3643	0.2350	0.5091	0.6554	0.7755
<code>section_only</code>	0.3520	0.2141	0.4987	0.6554	0.7885
<code>abstract_only</code>	0.3687	0.2480	0.4935	0.6606	0.7676
<code>full_rich</code>	0.3640	0.2350	0.5039	0.6423	0.7650
<i>PeerQA MiniLM+title (published)</i>	0.3654	—	—	—	—

*Replicates the PeerQA published BM25 template.

[†]Evaluated against paragraph-membership gold; see Section 9.5.

At paragraph level, `full_rich` (MRR 0.4036) and `abstract_only` (0.4027) are effectively tied and both improve over `title_only` by $\approx +1.6$ MRR points. The rank ordering of formats is informative: `full_rich` $\approx$ `abstract_only` $>$ `title+section+pos` $>$ `section_only` $\approx$ `title+section` $>$ `title_only` $>$ `plain`. The abstract’s first sentence is the dominant component: removing it (giving `title+section+pos`) drops MRR by ≈ 0.012 , while removing everything else (giving `abstract_only`) preserves the gain.

At sentence level, the winner *inverts*: `abstract_only` (MRR 0.3687) beats `full_rich` (0.3640) by 0.0047. Adding `title + section + position` on top of the abstract re-introduces token overwhelming—too much context relative to a short sentence dilutes the sentence’s keyword signal. Position tags help short units disproportionately ($+0.0108$ at sentence level vs. $+0.0046$ at paragraph level), because a sentence’s vocabulary is sparse enough that a structural cue like “Para 7 Sent 3” provides meaningful disambiguation.

9 Analysis

9.1 Cross-Approach Comparison

Table 11 places all results in context alongside the published PeerQA baselines.

Table 11: Cross-Approach Paragraph-Level MRR Summary (n=383)

Method	MRR	Source
Zero-shot MiniLM (Approach 2 base)	0.3286	This work
Contriever zero-shot	0.3494	PeerQA paper
Supervised MiniLM (Approach 2)	0.3618	This work
BM25 <code>title_only</code> baseline	0.3873	This work
BM25 <code>full_rich</code> (Approach 3)	0.4036	This work
BM25 published baseline	0.4288	PeerQA paper
BM25 + Query2Doc (Approach 1)	0.4432	This work
Dragon+	0.4845	PeerQA paper

Approach 1 is the strongest single result, surpassing the published BM25 baseline by $+0.0144$ MRR. Approach 3 is the most cost-efficient, reaching MRR 0.4036 with no API calls and 50 seconds of CPU runtime. Approach 2 is the only method operating on learned embeddings and shows the most room for improvement with a larger base model, more training epochs, or GPU-scale hard negative mining.

9.2 Why Query2Doc Outperforms Paraphrasing

Paraphrasing and Query2Doc attack the lexical gap from opposite sides. Paraphrasing rewrites the *question*: it substitutes synonyms but keeps the vocabulary in the register of a question. Query2Doc generates the *answer*: it produces formal scientific prose with technical terminology that would appear in the paper. BM25 is a bag-of-words matcher, so new discriminative terms in the pseudo-document (e.g., “convergence rates,” “attention scores,” “ablation study”) directly increase overlap with the evidence paragraph. A paraphrase that says “utilization” for “use” adds nothing; a pseudo-document that adds “cross-validation,” “hyperparameter tuning,” and “validation loss” to a question about model training adds several new matching signals.

9.3 Why RRF Hurts in Approach 1

With Query2Doc at MRR 0.4432 and original BM25 at 0.3852, RRF fusion yields MRR 0.4174—pulled down ≈ 0.026 . RRF earns its keep only when component rankers are competitive and make complementary errors [2]. When one is dominant, fusion introduces noise from the weaker ranker into the aggregation, degrading the strong component’s placement of the relevant document. The practical implication is to use the strong ranker alone when one strategy is decisively dominating.

9.4 Why Same-Paper Hard Negatives Matter

Random corpus negatives from unrelated papers share neither domain vocabulary nor topic, making the contrastive task trivially easy and the model would only learn coarse domain boundaries. By sampling negatives from the *same paper*, each training example forces the model to distinguish between topically related paragraphs, which is the actual difficulty of PeerQA retrieval. A question about a paper’s method must be separated from that same paper’s results, background, and limitations paragraphs, all written in identical technical vocabulary. Same-paper negatives directly train the model for this fine-grained intra-document discrimination.

9.5 Granularity-Dependent Optimal Contextualization

The most striking finding of Approach 3 is that the winning format differs by retrieval granularity. At paragraph level, more anchors help: ~ 338 characters of paragraph text is long enough to maintain its BM25 signal even with a multi-component prefix. At sentence level, only the *single most informative* anchor—the abstract’s first sentence—should be used; adding more components re-introduces token overwhelming. Position tags are most beneficial where the text is shortest: structural disambiguation adds a high-signal cue when the evidence unit’s own vocabulary is sparse. This finding provides actionable design guidance: match the richness of the context template to the length of the retrieval unit.

9.6 Why Approach 1 Outperforms Approach 3

Query2Doc (Approach 1, MRR 0.4432) outperforms even the best structural contextualization format (Approach 3 *full_rich*, MRR 0.4036) by a margin of $+0.0396$ MRR. This is because the two approaches attack different gaps: Approach 3 improves document

representations using existing metadata already present in the corpus (abstract, title, section heading), while Approach 1 generates *new* technical vocabulary in the query by predicting the answer. Approach 3 is constrained to information already in the index; Approach 1 leverages LLM parametric knowledge to anticipate vocabulary that BM25 would otherwise miss. The two approaches are complementary in principle, and combining them—using *full_rich* document representations with Query2Doc-augmented queries—is a natural direction for future work.

9.7 Complementary Error Profiles

The three approaches are complementary by design. Approach 1 closes the *lexical gap* on the query side through BM25 term matching on an enriched query. Approach 2 closes the *semantic gap* via learned embeddings that generalize beyond exact-term overlap. Approach 3 closes the *structural anchoring gap* on the document side by adding metadata that orients BM25 without modifying the query or model. A natural next step is to stack all three: use Approach 3’s *full_rich* document representations with Approach 1’s Query2Doc queries, then re-rank with an Approach 2-style fine-tuned dense retriever as a second stage. Each component addresses a different failure mode and their errors should be largely non-overlapping.

10 Downstream Generation Evaluation

10.1 Setup

Using the top-10 paragraphs retrieved by Approach 1 (Query2Doc BM25, $R@10 = 0.6527$) as context, we generate free-form answers with *gpt-4o-mini* as a CPU-accessible proxy for Llama-3-8B-Instruct [4], using the same RAG prompt structure from the PeerQA paper. We evaluate on the 245 questions that have a non-null expert-written reference answer. Generation quality is measured with Rouge-L F1 and an AlignScore proxy (sentence-level NLI entailment via cross-encoder/*nli-deberta-v3-small*). For reference, the published PeerQA generation baseline uses SPLADEv3 [8] as its retriever with the same RAG-10 setup.

10.2 Results

Table 12: Downstream Generation Results (Query2Doc RAG-10, $n=245$)

Metric	Value
Rouge-L F1	0.1686 ± 0.1086
Rouge-L Precision	0.1529
Rouge-L Recall	0.2759
AlignScore proxy	0.0719 ± 0.1666

A mean Rouge-L F1 of 0.1686 is consistent with the expected range for open-ended scientific QA. The recall-heavy breakdown (recall 0.2759 vs. precision 0.1529) reflects the typical behavior of instruction-tuned LLMs, which generate complete explanatory sentences even when the gold reference is a terse fragment.

The score distribution is bimodal: questions where retrieval succeeds ($R@10 = 1$, $\sim 65\%$ of questions) reach Rouge-L values up to

0.692 when the generator faithfully reproduces specific technical values from the retrieved context, while questions where retrieval fails produce near-zero scores from either generic hedging or factual hallucination from parametric memory. This confirms that retrieval quality, not generator capacity, is the primary bottleneck for this task.

The AlignScore proxy mean of 0.0719 does not reflect actual factual consistency; it reflects a calibration failure of the small NLI cross-encoder on long-context inputs. When all 10 retrieved paragraphs (~1,000–2,000 tokens) are concatenated as a single premise, the model’s 512-token positional encoding limit is exceeded and output logits collapse toward the neutral/contradiction classes regardless of actual entailment. Valid factual consistency evaluation requires the full AlignScore-large.ckpt (RoBERTa-large, 355M parameters) or a sentence-level truncation strategy.

11 Conclusion

We evaluated three evidence retrieval approaches on the full standard 383-question PeerQA evaluation set. We found that Query2Doc-augmented BM25 (Approach 1) is the strongest single approach, achieving MRR = 0.4432 and surpassing the published PeerQA BM25 baseline by +1.44 absolute MRR. The key lesson is that generating a hypothetical answer passage is more effective than paraphrasing the question, since the pseudo-document introduces new document-vocabulary terms that directly improve BM25 matching. Richer structural contextualization (Approach 3) achieves MRR = 0.4036 at zero cost, confirming that the abstract’s first sentence is the single most valuable context component and that the optimal template format is granularity-dependent—a finding with practical implications for BM25-based retrieval system design. Supervised contrastive fine-tuning (Approach 2) reaches MRR = 0.3618, outperforming the published zero-shot Contriever baseline and demonstrating that even a constrained single-epoch fine-tuning on ~440 training pairs provides a consistent, meaningful signal. The downstream generation experiment (Rouge-L F1 = 0.1686) reinforces that retrieval quality is the primary bottleneck: when the correct evidence paragraph is retrieved, the generator faithfully reproduces specific technical values; when retrieval fails, no amount of generation capacity recovers the answer.

Future work should explore combining all three approaches in a unified pipeline—using Query2Doc query enrichment over full_rich document representations, re-ranked by a domain-adapted dense retriever with GPU-scale training—and evaluate downstream generation quality with the full AlignScore model.

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